AI-GENERATED IMAGE DETECTION WITH POST-PROCESSING ROBUSTNESS ANALYSIS

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ABSTRACT

In today's world there are large number of AI generated images circulating all over the internet, and many of them are so realistic that it is very hard to distinguish between which one is real image and which one is AI generated. Therefore, problems associated with misinformation, image forgery, and media authenticity arise. For this purpose, a novel classification scheme based on transfer learning of a pretrained ResNet50 model on the CIFAKE dataset is proposed. Apart from image classification itself, robustness of the classifier is studied utilizing an optimization procedure that makes modifications to fake AI images in order to minimize the model's confidence of predicting those images as fakes while maintaining their structure. The experiments reveal the critical importance of the model's robustness to even minor modifications.

***Index Terms*—** AI-generated image detection, deep learning, ResNet50, adversarial robustness, CIFAKE dataset

1. INTRODUCTION

With the rapid growth of generative AI, many researchers have worked on detecting AI-generated images using deep learning techniques. Different approaches have been proposed to identify subtle patterns and artifacts present in synthetic images that are often difficult for humans to notice. Dhakal et al. introduced SimLBR: Learning to Detect Fake Images by Learning to Detect Real Images, where the focus was placed on learning the characteristics of real images instead of directly learning fake-image artifacts. Their work showed that understanding real-image distributions can improve the generalization ability of fake-image detectors. The paper Level Up the Deepfake Detection explored the use of deep learning models for detecting GAN-generated and diffusion-generated images. The authors demonstrated that convolutional neural networks can effectively capture hidden synthetic patterns and achieve high detection accuracy across different datasets. In It Until You Break It researchers looked at how Fake It Until You Break It researchers can trick artificial intelligence image detectors. They found out that Fake It Until You Break It researchers can make these detectors get it wrong by making changes to the image. These small changes can make a difference in how sure the detectors are about what they see even if the image still looks a lot like the original image, to our eyes. Another study titled Vulnerabilities in AI-generated Image Detection further analyzed the robustness limitations of AI-image detectors. The authors showed that optimized perturbations can manipulate model predictions and transfer successfully across different detection models. The work Towards Robust GAN-generated Image Detection focused on improving the robustness and generalization capability of synthetic image detectors. The study emphasized the importance of learning stronger feature representations capable of handling unseen AI-generated images and adversarial modifications. A related study, Detecting AI-Generated Images Using a Hybrid ResNet-SE Architecture, used transfer learning with ResNet-based architectures for classifying real and fake images. Their work demonstrated that pretrained deep learning models can achieve strong classification performance on datasets similar to CIFAKE. Finally, the RAID benchmark paper presented a large-scale evaluation framework for testing the robustness of AI-image detection systems under adversarial conditions. The benchmark highlighted several weaknesses of current detection models and emphasized the need for robustness-aware evaluation techniques. Although many existing works focus on improving classification accuracy, limited research has explored optimization-based image modification for robustness analysis. The proposed work extends previous studies by combining AI-image detection with constrained image optimization to analyze how small, visually preserved perturbations can influence model predictions.

2. RELATED WORK

With the growing popularity of generative AI models, the AI-generated images are becoming increasingly photorealistic. Despite the usefulness of these technologies in many different domains, there are also risks connected with their use, such as misleading information, creation of fake media and digital manipulation. As a result, there is a need for develop-

ing a way to identify AI-generated images. For this reason, this research proposes a deep learning-based model based on ResNet50 pretrained model for classification of real and AI-generated images from the CIFAKE dataset. Additionally, the investigation explores the stability of the AI-image detector by making slight changes to AI-generated images using an optimization-based method. Specifically, the task is to decrease the detector's belief about an image being artificial while not compromising its structural integrity using certain regularization techniques. This investigation demonstrates that small changes made to the AI-generated images have a significant impact on the prediction process of the AI-image detector.

3. METHODOLOGY

The proposed methodology consists of two main stages: AI-image detection and optimization-based image modification. First, a deep learning model is trained to classify real and AI-generated images using transfer learning with ResNet50. After detection, AI-generated images identified by the model are iteratively modified using an optimization approach to reduce the model's fake prediction confidence while preserving the original visual structure of the image.

3.1. Dataset

The CIFAKE dataset was used to train and test the model. This CIFAKE dataset has pictures and pictures that were made by a computer to look real and they are sorted into different folders so the model can tell them apart. The CIFAKE dataset has a lot of pictures. 100,000 In the group used for training and 20,000, in the group used for testing. We picked the CIFAKE dataset because it has both pictures and computer made pictures, which is what we needed.

3.2. Image Processing

Before training, all images were resized to 224×224 pixels to match the input requirements of the ResNet50 model. The images were then converted into tensor format for numerical processing in PyTorch. Normalization was also applied using ImageNet mean and standard deviation values to maintain consistency with the pretrained ResNet50 architecture.

3.3. AI Image Detection Model

A pretrained ResNet50 model was used for image classification through transfer learning. Initially, most of the pretrained layers were frozen to preserve previously learned visual features such as edges, textures, and patterns. The final fully connected layer was replaced with a two-class output layer for classifying images as REAL or FAKE. The model was trained using the Adam optimizer and cross-entropy loss function.

3.4. Optimization-Based Image Modification

After classification, AI-generated images detected by the model were further processed using an optimization-based modification approach. Instead of changing the model parameters, the image itself was iteratively modified to reduce the model's confidence in predicting it as fake. A combined loss function consisting of classification loss and regularization loss was used during optimization. The classification loss pushed the prediction toward the REAL class, while the regularization term helped preserve the structural similarity between the original and modified image. This approach was used to analyze the robustness and limitations of AI-image detection systems under constrained image modifications.

4. EXPERMINATAL RESULTS

The model we made using the ResNet50 architecture was tested on the CIFAKE data set. We used NVIDIA RTX 4050 Laptop GPU to make it go faster. The results of the test showed that the model was right about the pictures it was trained on 97.76 percent of the time. It was also right about the pictures it was tested on 97.20 percent of the time. We looked at the confusion matrix and the reports on how the model did. The model was very good at telling the difference between FAKE images. It was about 0.97 percent of the time.

This means that the ResNet50 model we made is good, at finding the things in the pictures. The ResNet50 model learned a lot from the CIFAKE data set.

For robustness analysis, AI-generated images predicted by the model were further processed using the proposed optimization-based image modification approach. Initially, selected AI-generated images were classified as FAKE with very high confidence. During optimization, the classification loss decreased significantly while maintaining a low regularization loss, indicating that the image structure was preserved while reducing the model's confidence in identifying the image as fake.

Experimental observations showed that small, visually constrained modifications were capable of changing the model prediction from FAKE toward REAL without introducing major visual distortions. This demonstrates the sensitivity and robustness limitations of deep learning-based AI-image detection systems under constrained perturbations.

5. DISCUSSION

The experimental results show that the proposed ResNet50-based framework performs effectively in distinguishing real and AI-generated images. The high classification accuracy indicates that deep learning models are capable of learning subtle visual patterns and artifacts present in synthetic images. Transfer learning using a pretrained ResNet50 architecture also helped improve performance while reducing training

complexity and time.

The optimization-based image modification approach further demonstrated that even small and visually constrained changes can significantly influence the predictions of AI-image detection systems. By minimizing the classification loss while preserving image structure through regularization, the modified images were able to reduce the model's confidence in predicting them as fake. This highlights an important limitation of current AI-image detection systems and shows that such models may still be vulnerable to carefully optimized perturbations.

One limitation observed during experimentation was the low resolution of images in the CIFAKE dataset. Since the images were resized to 224×224 pixels for ResNet50, some outputs appeared visually blurred during comparison. However, this did not significantly affect the classification or optimization process. Future work can explore higher-resolution datasets and more advanced regularization methods to improve visual quality and robustness analysis.

6. CONCLUSION

In this work, a deep learning-based framework was proposed for detecting AI-generated and real images using transfer learning with ResNet50. The model achieved high classification accuracy on the CIFAKE dataset and successfully learned meaningful visual patterns useful for synthetic image detection.

In addition to classification, this work also explored the robustness of AI-image detectors through an optimization-based image modification approach. By iteratively modifying AI-generated images while preserving their structural similarity, the model's prediction confidence was significantly altered. The results demonstrate that even small constrained perturbations can influence deep learning-based detection systems.

Overall, the proposed work not only provides an effective AI-image detection framework but also highlights important robustness challenges that need to be considered in future synthetic media detection systems.

7. REFERENCES

[1] A. Dhakal, S. Khanal, S. Sastry, J. Arndt, P. A. Dias, D. Lunga, and N. Jacobs, "SimLBR: Learning to Detect Fake Images by Learning to Detect Real Images," arXiv preprint arXiv:2602.20412, 2026.

[2] L. Guarnera, O. Giudice, and S. Battiato, "Level Up the Deepfake Detection: A Method to Effectively Discriminate Images Generated by GAN Architectures and Diffusion Models," arXiv preprint arXiv:2303.00608, 2023.

[3] S. Mavali, J. Ricker, D. Pape, Y. Sharma, A. Fischer, and L. Schönherr, "Fake It Until You Break It: On the Adversarial Robustness of AI-generated Image Detectors," arXiv preprint arXiv:2410.01574, 2024.

[4] Y. Diao, N. Zhai, C. Miao, X. Yang, and M. Wang, "Vulnerabilities in AI-generated Image Detection: The Challenge of Adversarial Attacks," arXiv preprint arXiv:2407.20836, 2024.

[5] C. Liu, T. Zhu, S. Shen, and W. Zhou, "Towards Robust GAN-generated Image Detection: A Multi-view Completion Representation," arXiv preprint arXiv:2306.01364, 2023.

[6] A. R. Gunukula, H. Das Gupta, and V. S. Sheng, "Detecting AI-Generated Images Using a Hybrid ResNet-SE Attention Model," Applied Sciences, vol. 15, no. 13, Art. no. 7421, 2025.

[7] H. Eddoubi, J. Ricker, F. Cocchi, L. Baraldi, A. Sotgiu, M. Pintor, M. Cornia, L. Baraldi, A. Fischer, R. Cucchiara, and B. Biggio, "RAID: A Dataset for Testing the Adversarial Robustness of AI-Generated Image Detectors," arXiv preprint arXiv:2506.03988, 2025.